Retail Revenue Optimization Strategy

A Data Driven Case Study

By Kashish Rathore

Summary

This case study is to identify key drivers of revenue of a fast fashion retailer.

Findings:

- Promotion effectiveness depends on placement.
- Mid and premium segments contribute most revenue
- Price matters more than as it drives more revenue than volume of sales.

Recommendations:

- Promotions should be used on targeted items.
- Focus on pricing strategy.

Problem Statement

Retail businesses decide how to price products, position them, and use promotions to maximize revenue.

This study aims to identify which factors most strongly influence revenue.

Approach

- Cleaned and prepared dataset.
- Created revenue metric
- Segmented products by price range
- Analysed impact of promotion and product placement

Insights

1. Revenue Distribution : is highly skewed, with a small number of products generating significantly higher revenue.

Interpretation:
A limited set of high performing products contribute to total revenue.
Business Implication:
Focus on identifying and prioritizing top revenue generating products through better visibility and inventory.

2. Promotion Effectiveness : Promoted products generate slightly higher revenue than non promoted.
Interpretation:
Promotion contributes to revenue uplift but is not a primary driver of performance.
Business Implication:
Use promotions selectively and combine them with other strategies such as product placement for maximum effectiveness.

3. Promotion × Product Position : Effect of Promotions depend on position of the items. Promotion has strong impact in low-visibility areas (End cap) but limited impact in high visibility placements (Front of store).
Interpretation: High visibility placements already drive results, despite promotions.
Business Implication: Promotions should be used at positions like end cap and aisle where it creates the most value.

4.Price and Sales volume - There is a very weak negative correlation (-0.08) between price and sales volume.
Interpretation: Sales volume doesn't depend on price alone and there are other factors which influence the behaviour.
Business Implication : Pricing strategies should be supported by promotion and placement rather than treated as an the only factor.

5. Price Segmentation Strategy:
The majority of products fall within the mid price segment, with very few luxury products.

Interpretation: The business is primarily positioned in the midmarket segment, balancing affordability and style.

Business Implication: Continue focusing on mid range products while using premium and luxury segments strategically for margin growth.

6. Price segment
Insight: Sales volume decreases as price increases, with mid-range products achieving the highest demand.
Interpretation: Midpriced products offer the best balance between affordability and perceived value.
Business Implication: Focus on mid range pricing to maintain stable demand.

7. Price Segment vs Revenue
Higher price segments generate significantly higher revenue per product despite lower sales volume.
Interpretation: Revenue is driven more by pricing power than by sales volume.
Business Implication: Leverage premium pricing strategies for revenue growth while using budget products for customer acquisition.

8. Overall Pricing Strategy
Different price segments serve different roles:
- Budget → high volume, low revenue
- Mid → high volume, balanced revenue
- Premium → moderate volume, high revenue
- Luxury → low volume, very high revenue

Interpretation:
The business operates a hybrid pricing strategy balancing volume and margin.
Business Implication:
Maintain a diversified pricing strategy across segments to optimize overall performance.

9. Correlation Insight : Revenue is more strongly influenced by price (0.73) than by sales volume (0.51).
Interpretation:
Increasing price has a greater impact on revenue than increasing demand.
Business Implication:
Prioritize pricing optimization as a key driver of revenue growth.

10. Seasonal Products
Seasonal products generate lower median revenue compared to non-seasonal products.
Interpretation:
Seasonal demand may be inconsistent or limited.
Business Implication:
Focus on evergreen products for stable revenue and avoid over-investment in seasonal inventory**.**

11. Dataset Limitation

- The dataset is heavily skewed toward men's products.
- Insights related to product segments may be biased due to data imbalance.
- Further analysis with a more balanced dataset is required before making category-level decisions.

Recommendations

- Mid and premium pricing segments should be focused for revenue growth
- Optimize pricing rather than relying solely on increasing sales volume
- Promotions should be based on product placement